

Data Encryption Standard (DES)

Aim

To implement the **Data Encryption Standard (DES)** algorithm for encrypting and decrypting a plaintext message using a secret key, thereby ensuring confidentiality of data.

Algorithm

Encryption Algorithm

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the secret key.

Step 4: Convert the plaintext into binary format.

Step 5: Apply the Initial Permutation (IP).

Step 6: Divide the data into two equal halves (Left and Right).

Step 7: Generate 16 round keys from the original key.

Step 8: Perform 16 rounds of DES operations:

- Expand the right half.
- XOR with the round key.
- Substitute using S-Boxes.
- Apply permutation.
- XOR with the left half.
- Swap the left and right halves.

Step 9: Combine both halves.

Step 10: Apply the Final Permutation (FP).

Step 11: Display the ciphertext.

Step 12: Repeat the same process in reverse order for decryption.

Step 13: Display the original plaintext.

Step 14: Stop the program.

C Program (Simplified DES Demonstration)

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```

char plaintext[100], key[100], ciphertext[100], decrypted[100];
int i, len;
printf("Enter Plaintext: ");
scanf("%s", plaintext);
printf("Enter Secret Key: ");
scanf("%s", key);
len = strlen(plaintext);
// Encryption using XOR (Simplified DES demonstration)
for(i = 0; i < len; i++)
{
    ciphertext[i] = plaintext[i] ^ key[i % strlen(key)];
}
ciphertext[len] = '\0';
printf("\nEncrypted Text: ");
for(i = 0; i < len; i++)
{
    printf("%02X", (unsigned char)ciphertext[i]);
}
// Decryption
for(i = 0; i < len; i++)
{
    decrypted[i] = ciphertext[i] ^ key[i % strlen(key)];
}
decrypted[len] = '\0';
printf("\nDecrypted Text: %s\n", decrypted);
return 0;
}

```

OUTPUT :

The image shows a screenshot of an IDE (Embarcadero Dev-C++ 6.3) with a C++ program for DES encryption and decryption. The code is in a file named DES.cpp. The program prompts the user to enter a plaintext and a secret key, then displays the encrypted and decrypted text. The output window shows the execution results.

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char plaintext[100], key[100], ciphertext[100], decrypted[100];
7     int i, len;
8
9     printf("Enter Plaintext: ");
10    scanf("%s", plaintext);
11
12    printf("Enter Secret Key: ");
13    scanf("%s", key);
14
15    len = strlen(plaintext);
16
17    // Encryption using XOR (Simplified DES demonstration)
18    for(i = 0; i < len; i++)
19    {
20        ciphertext[i] = plaintext[i] ^ key[i % strlen(key)];
21    }
22    ciphertext[len] = '\0';
23
24    printf("\nEncrypted Text: ");
25    for(i = 0; i < len; i++)
26    {
27        printf("%02X", (unsigned char)ciphertext[i]);
28    }
29
30    // Decryption
31    for(i = 0; i < len; i++)
32    {
33        decrypted[i] = ciphertext[i] ^ key[i % strlen(key)];
34    }
35    decrypted[len] = '\0';
36}
```

Compiler: 0 Errors, 0 Warnings, 0 Output Filesize: 323.2841796875 Kib, 0.42s Compilation Time

Output Window:

```
Enter Plaintext: HELLO
Enter Secret Key: KEY

Encrypted Text: 030015070A
Decrypted Text: HELLO

-----
Process exited after 6.168 seconds with return value 0
Press any key to continue . . .
```